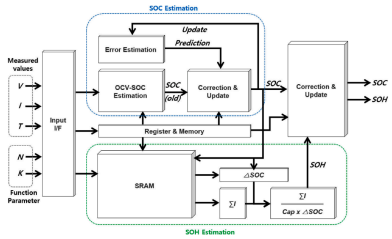
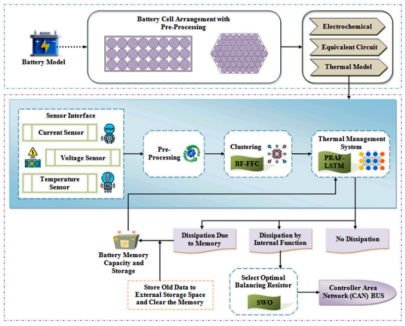
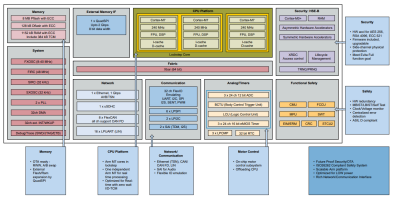
Sl no.	Category	Paper/thesis Title	Brief Description	Web URL
1	SoC	A comparative study and validation of state estimation algorithms for Li-ion batteries in battery management systems	**Table 1** Code properties of the C-code generated estimation algorithms. Algorithm / RAM (kB) / Stack-size (kB) / Complexity Luenberger observer / 2.16 / 0.05 / 5 Sliding-mode observer / 2.30 / 0.05 / 13 Extended Kalman Filter / 3.54 / 1.24 / 16 Sigma-point Kalman Filter / 3.67 / 1.25 / 38 Table 1 Code properties of the C-code generated estimation algorithms	https://www.sciencedirect.com/science/article/abs/pii/S0306261915007357
2	SoC	Optimized estimation of Li-ion battery parameters to improve the Enhanced Self-Correcting model with nonlinear least-squares data fitting and SoC estimation using Invariant EKF	Page 5 first paragraph of the pdf, according to the .map file generated by Simulink, the memory size of the code is 19.4 KB (.cinit-.econst-.text.1-.text.2) of flash memory and 3.2 KB (.stack-.ebss) of RAM.	https://www.sciencedirect.com/science/article/pii/S2772671125002220
3	SoC	Analysis and advancements of the state of charge estimation methods in smart battery management system supported by lithium-ion battery operated electric vehicles	Page 11-12 has (not directly memory analysis) various SoC estimation methods complexity analysis, advantages and disadvantages	https://www.sciencedirect.com/science/article/pii/S2949821X25001000
4	SoC and SoH	VLSI design and FPGA implementation of state-of-charge and state-of-health estimation for electric vehicle battery management systems	 Fig. 7. Logic Circuit Design of the proposed SOC and SOH estimation block. FPGA resource utilization report for a design implemented on a Xilinx Zynq-7020 FPGA	https://www.sciencedirect.com/science/article/pii/S2352152X23022739
5	Battery balancing	Analysis of battery lifespan improvement with SOC battery	Page 7 second column first paragraph, when deploying the SOC estimation and	https://www.sciencedirect.com/science/article/pii/S

		balancing technique	terminal voltage-based battery balancing techniques on an MCU, the SOC-based method uses around 50.34 kB of memory. In contrast, the terminal voltage-based method uses 10.54 kB.	2352152X25032566
6	Thermal Management	A computational analysis based on thermal management system for electric vehicle battery packs with different cell arrangements	 Fig. 1. Block diagram for the proposed research model. This is the research model that states that when memory is full (dissipation due to memory), store the old data to external storage space and clear the memory.	https://www.sciencedirect.com/science/article/pii/S0735193325012588

Miscellaneous (initial research)

Sl no.	Paper/thesis Title	Brief Description	Web URL
1	A comparative study and validation of state estimation algorithms for Li-ion batteries in battery management systems	The results show that a simple estimation method like the sliding-mode observer can compete with the Kalman-based methods presenting less computational time and memory usage.	https://www.sciencedirect.com/science/article/abs/pii/S0306261915007357
2	Analysis and development of Battery Management System for electric vehicles	The firmware structure and its layers, DMA, and UART	https://webthesis.biblio.polito.it/22767/1/tesi.pdf
3	Estimation of Essential Battery State Parameters for Battery Management Systems (BMS) in Electric Vehicles Using Long Short Term Memory (LSTM) and Time Series Analysis Combined With Extended Kalman Filter	LSTM and its Section A and E has informative tables, i.e., "Comparison of various models with respect to runtime and memory requirement" and "Comparison of runtime for various algorithms with different memory availabilities."	https://www.researchgate.net/publication/394780391_Estimation_of_essential_battery_state_parameters_for_battery_management_systems_BMS_in_Electric_Vehicles_using_l

			ong_short_term_memory_LSTM_and_time_series_analysis_combined_with_Extended_Kalman_Filter
4	Choice of the Data Logging Memory for Battery Management Systems	Table 1 contains comparison of typical non-volatile memories and F-RAM	https://link.springer.com/article/10.1007/s38314-021-0665-0
5	Battery Management System for Electric Vehicles: Comprehensive Review of Circuitry Configuration and Algorithms	Memory classifications and characteristics on page 21 of the pdf	https://www.mdpi.com/2032-6653/16/8/451
6	S32K3xx Data Sheet (Rev. 13 — 12 November 2025)	 Figure 13. S32K358 ASIL D Lockstep Core + One, 8MB General Purpose MCU This image contains S32K358 (the BMU board) block diagram	https://www.nxp.com/docs/en/data-sheet/S32K3xx.pdf
7	Recording frequency optimization for massive battery data storage in battery management systems	This research paper proposes a multi-frequency data logging method for EV battery systems that optimizes memory usage by dynamically adjusting signal recording rates based on signal variability, using wavelet and Fourier analysis to balance data accuracy and storage constraints.	https://www.sciencedirect.com/science/article/abs/pii/S0306261916312491?casa_token=wbQBD0iqBn8AAAAA:JROyDvr3rwSbISSAcRGXad12WAUnI5QZ32nw93sLpJmXao6Z2HPUehwl_S1CUJwhuQ7V-fv8PQ
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